

Lecture 29

Bee-Inspired Search and Swarm Robotics

COMP532 – Machine Learning and Bioinspired Optimisation

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Lecture goals

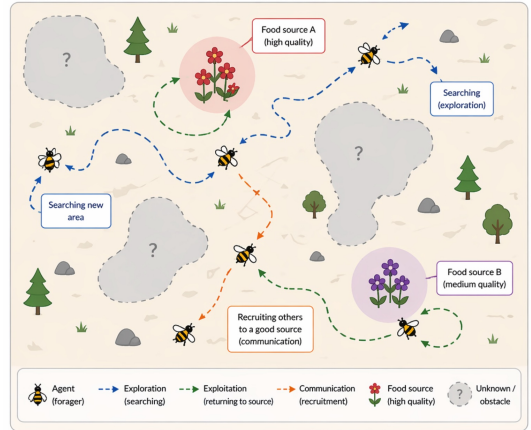
- Introduce **bee-inspired search** as a distributed approach to exploration, recruitment, and resource allocation.
- Explain how **path integration**, **direct communication**, and **division of labour** support collective foraging.
- Show how swarm ideas extend from optimisation algorithms to **embodied multi-robot systems**.
- Summarise the broader strengths of swarm methods: **scalability**, **robustness**, and **adaptivity**.

The foraging problem

- A group must locate and collect useful items in an unknown environment.
- The environment may be large, partially observable, and dynamic.
- Agents must decide where to search, when to exploit a known source, and how to coordinate effort.

Why it matters

Foraging captures core elements of routing, patrolling, transportation, exploration, and task allocation.



Why bees are a useful model

- Bees combine **distributed exploration** with **direct recruitment**.
- They do not need a central planner to allocate effort.
- Better food sources naturally attract more attention through communication.
- This gives a compelling model of **adaptive collective search**.

Core idea

A successful swarm must not only discover good opportunities; it must also **reallocate population effort** towards them.

1. Path construction / planning

- Where should an agent search?
- How should it move in an unknown space?
- How can new targets be discovered efficiently?

2. Path exploitation / repair

- Once a useful resource is known, how should effort be concentrated?
- How should routes be reused, improved, or repaired?
- How does the group adapt if conditions change?

Bee-inspired search ingredients

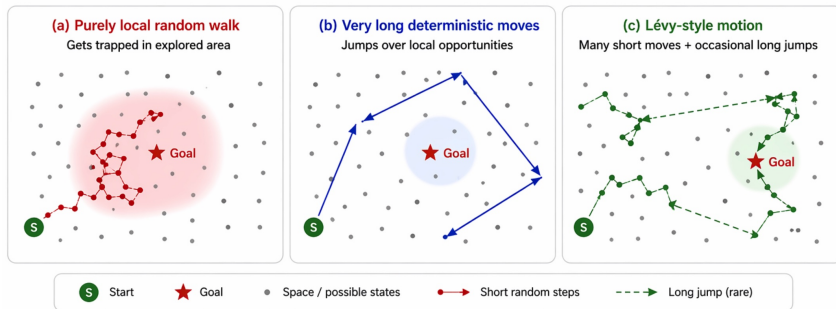
- **Lévy flight** for exploration: occasional long moves mixed with many short moves.
- **Path integration** for navigation: maintain an internal estimate of direction and distance.
- **Recruitment** for coordination: communicate promising food sources to others.

Interpretation

Bee-inspired systems combine **exploration**, **memory**, and **communication** in a distributed way.

Lévy flight intuition

- Purely local random walks can waste time in already explored regions.
- Very long deterministic moves can miss local opportunities.
- Lévy-style motion mixes many short searches with occasional long jumps.



Why this helps

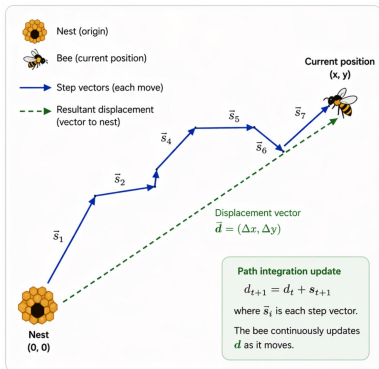
It supports both **thorough local search** and **broad spatial coverage**.

Path integration

- A bee can keep track of its displacement relative to the nest.
- This displacement can be represented as a vector. For example, if the bee is 3 m east and 4 m north of the nest, its path-integration vector is $\mathbf{v} = (3, 4)$. To return directly, it should move along the opposite vector $-\mathbf{v} = (-3, -4)$. The vector provides a compact form of navigational memory.

Key benefit

An agent can wander stochastically, yet still retain a direct estimate of how to return home.



Recruitment and the dance metaphor

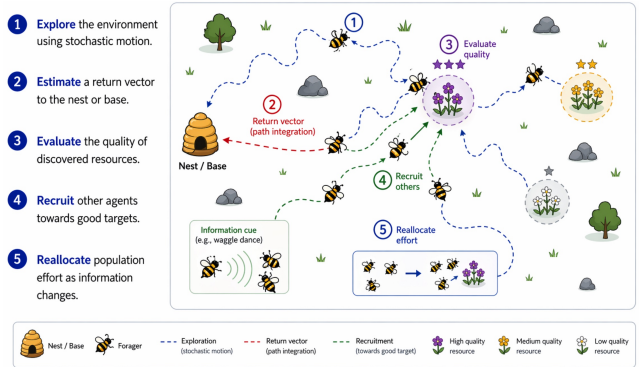
- After discovering food, a bee returns to the hive.
- It communicates information about direction, distance, and quality.
- Better sources generate stronger recruitment.
- The colony therefore reallocates effort through local communication.

Important point

Unlike ants, bees rely more on **direct communication** than on persistent environmental traces.

Bee-inspired foraging as a search template

1. Explore the environment using stochastic motion.
2. Estimate a return vector to the nest or base.
3. Evaluate the quality of discovered resources.
4. Recruit other agents towards good targets.
5. Reallocate population effort as information changes.



Comparing ants and bees

Ant-inspired systems

- indirect communication
- pheromone trails
- environmental memory
- often strong for path reuse

Bee-inspired systems

- more direct communication
- vector-based memory
- recruitment at the nest
- often strong for adaptive exploration

From swarm algorithms to swarm robotics

- Real robots operate with local sensors, noisy perception, and communication limits.
- Centralised control may be brittle, expensive, or impossible.
- Swarm principles offer a natural way to coordinate many simple robots.

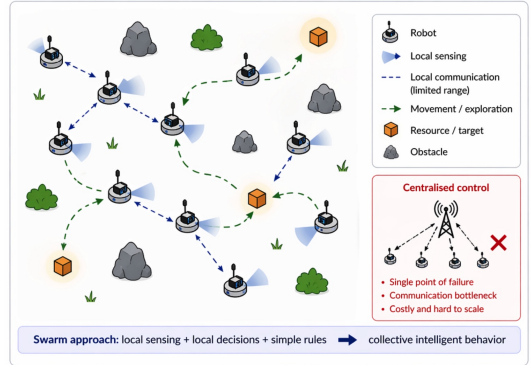


Interpretation

Swarm robotics is not only biologically inspired; it is also an engineering response to **distributed constraints**.

Why swarm robotics?

- **Limited communication:** robots may only communicate locally or intermittently.
- **Scalability:** adding more robots should not require redesigning a central controller.
- **Robustness:** failure of a few robots should not collapse the whole system.
- **Adaptivity:** the group should respond to changes in the environment.



Example: multi-robot foraging

- Multiple robots explore for resources with limited visibility.
- They communicate when they encounter each other.
- Path integration and visual markers can help identify nest and food.
- Recruitment-like information sharing improves collective efficiency.

Engineering translation

Bee-inspired principles can be implemented as practical coordination mechanisms in robot teams.

<https://www.youtube.com/watch?v=RV4l-UHRMrA>

Common swarm robotics application classes

- exploration and area coverage
- anomaly detection and monitoring
- patrolling and surveillance
- cooperative transport and task allocation
- aerial or mobile sensor swarms

Unifying feature

These settings reward systems that can coordinate effectively under **partial information** and **local interaction constraints**.

Design principles for swarm systems

- Keep local rules simple and reliable.
- Use randomness to preserve exploration.
- Add lightweight memory so useful discoveries are not lost.
- Prefer graceful degradation over brittle optimality.
- Design communication to be sparse, local, and informative.

Strengths of swarm methods

- decentralised operation
- parallel search
- robustness to failure
- flexibility under changing conditions
- strong match to multi-agent and robotic settings

Limitations and cautions

- local rules are simple, but system-level behaviour can still be hard to predict
- convergence may be slow or noisy
- coordination quality can depend strongly on sensing and communication assumptions
- distributed systems trade global precision for robustness and scalability

Takeaway

Swarm methods are powerful when centralised optimisation is unrealistic, but they are not magic replacements for careful system design.

Final takeaway

Closing message

The deepest lesson of swarm intelligence is not about insects. It is about how simple local mechanisms can generate effective global search and coordination in systems too large, noisy, or distributed for central control.